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Mesozoic paleomagnetism and northward translation of the Baja California Peninsula

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ABSTRACT

Paleomagnetic data from Cretaceous igneous and sedimentary rocks of the Peninsular Ranges and Continental Borderland of the Baja California Peninsula yield an average pole position of 83°N, 344°E (A_{95} = 3.4°), suggesting that the peninsula as a whole has moved northward ~11° and rotated 32° clockwise, with respect to the North American craton, since Cretaceous time. Only a small portion of this movement is due to the Neogene opening of the Gulf of California. Results from stratified rocks giving both positive and negative fold tests are included in the average Cretaceous pole. The rocks that fail fold tests have been deformed and remagnetized by nearby intrusives that are only slightly younger; uncorrected directions from these sites closely match corrected directions from sites where the fold tests are positive. Oceanic rocks were also sampled in the Continental Borderland from Upper Triassic and Upper Jurassic tectonostratigraphic terranes which are apparently allochthonous with respect to the western Baja Peninsula and each other. These oceanic rocks show magnetizations indicating low paleolatitudes, although the hemisphere of origin is presently unknown. The Cretaceous rocks of the Baja California Peninsula may have originally formed adjacent to the now-truncated continental margin along the northern Middle America Trench. Northward transport of the peninsula probably occurred by transcurrent faulting associated with oblique subduction of the Farallon plate beneath western North America between Late Cretaceous and late Miocene time.

INTRODUCTION

The California continental margin has undergone major tectonic changes, the most recent being the shift from a convergent, subduction-related margin to one marked by transform faulting. Motion along an early offshore transform fault began ~28 m.y. ago when the North American plate first overrode the East Pacific Rise and contacted the Pacific plate (Atwater and Molnar, 1973; Engebretson, 1983). With continued convergence, the transform lengthened until ~5.5 m.y. ago. At this time, the boundary jumped eastward to the incipient Gulf of California and present San Andreas fault system, attaching the Baja California Peninsula to the Pacific plate (Fig. 1). Since then, displacement between the Pacific and North American plates at an approximately constant rate of 5.6 cm/yr has moved the Baja Peninsula and a portion of southwestern California ~300 km northward relative to North America (Curray and Moore, 1984).

Although cumulative Neogene right-lateral offset along the northern portion of the San Andreas fault totals ~315 km (Nilsen and Clarke, 1975), metamorphic and Cretaceous granitic rocks of the Salinian terrane west of the San Andreas fault appear to have been displaced at least 500 km northward from inferred counterparts in southern California (Page, 1981; Fig. 1). Alternatively, recent paleomagnetic studies of Cretaceous and early Tertiary rocks in the Salinian and adjacent terranes (McWilliams and Howell, 1982; Champion and others, 1984) suggest that these terranes originated as much as 2,500 km south of their present location. Paleomagnetic results from many other Cordilleran terranes (Coney and others, 1980) indicate large northward translations and clockwise rotations relative to cratonic North America (Beck, 1976, 1980). These data imply that, prior to the development of the San Andreas transform, major northward tectonic transport of crustal fragments occurred along the western margin of North America during much of Tertiary time.

Many of the Mesozoic batholiths along the western edge of North America yield paleomagnetic data suggestive of origins far south of their present positions (Beck and others, 1981). Although paleomagnetic data from the plutonic rocks of the inboard Sierra Nevada are consistent with, or perhaps only slightly discordant with, data from rocks of stable North America (Grommé and others, 1967; Frei and others, 1984; Fig. 1), plutons of the Peninsular Ranges batholith near the continental margin in southern California appear to have moved northward ~11.5° and rotated 26° clockwise since the Cretaceous Period (Teissere and Beck, 1973; Erskine and Marshall, 1980). These data imply that the rocks of the Baja Peninsula farther to the south also may have moved much farther northward than the 300-km displacement associated with the Pliocene opening of the Gulf of California (Beck and Plumley, 1979). Paleomagnetic results from the Cretaceous Valle Formation on the Vizcaino Peninsula (Fig. 1) are consistent with this inferred northward translation of the Baja Peninsula (Patterson, 1984).

REGIONAL SETTING

The prebatholithic rocks of the Baja California Peninsula are predominantly Mesozoic in age and lie west of a recognizable Precambrian craton boundary that parallels the Gulf of California in western Sonora (Gastil and others, 1978; Fig. 1). These rocks comprise a number of tectonostratigraphic terranes. The easternmost Cortez terrane of the Baja Peninsula is probably composite and contains rocks of Mesozoic and both early and late Paleozoic ages (Gastil and Miller, 1981; Coney and Campa, 1984). To the west, the Santa Ana terrane (Coney and Campa, 1984) consists of

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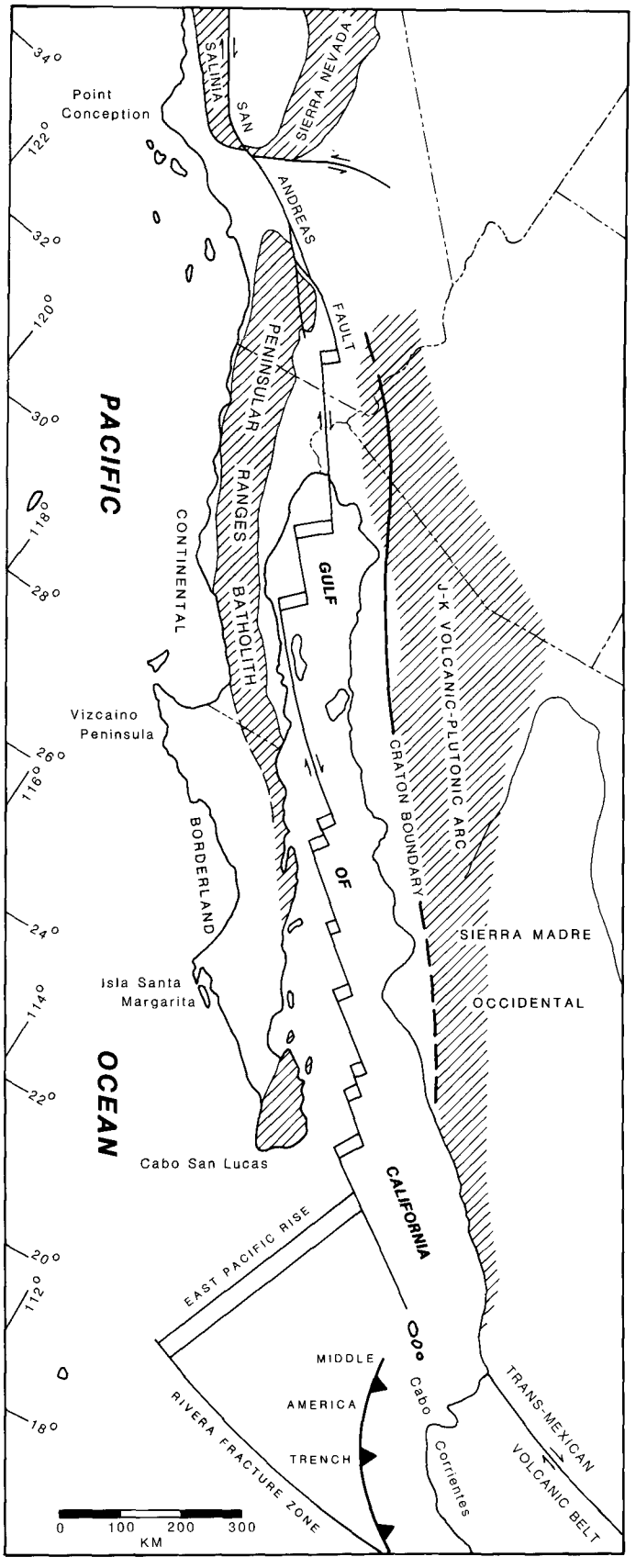


Figure 1. Regional setting of the Baja California Peninsula. Shaded areas indicate Mesozoic plutonic rocks. After Gastil and Jensky (1973), Gastil and others (1975), Karig and others (1978), and Gastil and Miller (1984).

Lower Jurassic through Lower Cretaceous andesitic volcanic and volcaniclastic sequences (including the Alisitos Formation) which host the batholithic rocks forming the core of the peninsula (Fig. 1).

Plutonic rocks of the Peninsular Ranges batholith were first emplaced in the Baja Peninsula in latest Jurassic and Early Cretaceous time, contemporaneously with volcanic-plutonic activity along a second continental arc in west-central Sonora and southern Arizona that had begun in Early Jurassic time (Gastil and others, 1981; Fig. 1). The Peninsular Ranges plutonic activity culminated by the middle Cretaceous, causing uplift, erosion, and westward-directed sedimentation that resulted in widespread deposition of the Valle Formation in the Continental Borderland (Patterson, 1979).

On the Vizcaino Peninsula, the Valle Formation overlaps the westernmost terrane of the Baja California Peninsula, the composite Vizcaino terrane (Coney and others, 1980), which contains several smaller oceanic terranes (Moore, 1984). Of the three principal smaller terranes, the structurally lowest (of unknown age) includes disrupted ophiolitic and arc-related blocks that have been subjected to middle Mesozoic low-temperature and high-pressure metamorphism in a serpentine-matrix mélange (Moore, 1984). The northern of the two overlying terranes is characterized by a basal Upper Triassic ophiolite and Upper Jurassic to Lower Cretaceous volcanogenic conglomerates, graywackes, and shales that contain intercalated volcanic rocks (the Eugenia Formation). The terrane to the south is juxtaposed with the northern terrane along the Sierra Placeres mélange and is characterized by a sequence consisting of Triassic ophiolite, Upper Triassic tuffaceous chert and limestone, and quartz-poor volcaniclastic sedimentary rocks of Early Jurassic age (the San Hipolito Formation). These rocks are intruded by Middle Jurassic to Lower Cretaceous tonalite plutons (Moore, 1984). The oceanic terranes of the Vizcaino Peninsula were probably sutured and accreted to the Baja Peninsula by the Early Cretaceous (Moore, 1983) and are separated from the Cretaceous volcanic and plutonic core of the Baja Peninsula by a major structural discontinuity that lies beneath the Quaternary fill of the Vizcaino basin (Rangin, 1978; Gastil and others, 1978; Fig. 2D).

Recent tectonic models of Gastil and others (1978, 1981) and Rangin (1978) suggest that two major, continentally related arc segments, one to the east of the craton boundary in western Mexico and the other straddling the continental margin west of the craton boundary (Peninsular Ranges batholith), occurred simultaneously along southwestern North America, at least during latest Jurassic and earliest Cretaceous time (Fig. 1). Alternate paleogeographic reconstructions of Alta California and the Baja California Peninsula, however, realign the Mesozoic batholithic and related geologic terranes to conform with a single subduction complex along the western edge of North America and are supported by the available paleomagnetic and geologic data (Hamilton, 1969; Jones and others, 1976; Gastil and others, 1972; Gastil and Miller, 1983). This paleomagnetic study of the Baja California Peninsula was undertaken to resolve these two tectonic models by determining more completely the peninsula's pre-Neogene displacement history with respect to the North American craton, as well as latitudinal origins for the oceanic rocks of the Vizcaino terrane.

FIELD AND LABORATORY PROCEDURES

Samples for paleomagnetic analysis were taken as 2.5-cm drill cores, using a portable gasoline-powered drill, and were oriented in the field to an estimated accuracy of 2°, using magnetic and solar compasses and a clinometer. Sampling localities (Fig. 2) were restricted to both artificial exposures and to the freshest natural exposures such as stream-bank cuts and wave-cut surfaces. Wherever possible, samples were collected from localities with variable attitudes to test the age of the rock's magnetization with respect to the age of folding or tilting. Structural corrections for all magnetization directions were accomplished by a simple rotation of the measured bedding plane into the horizontal about the strike axis.

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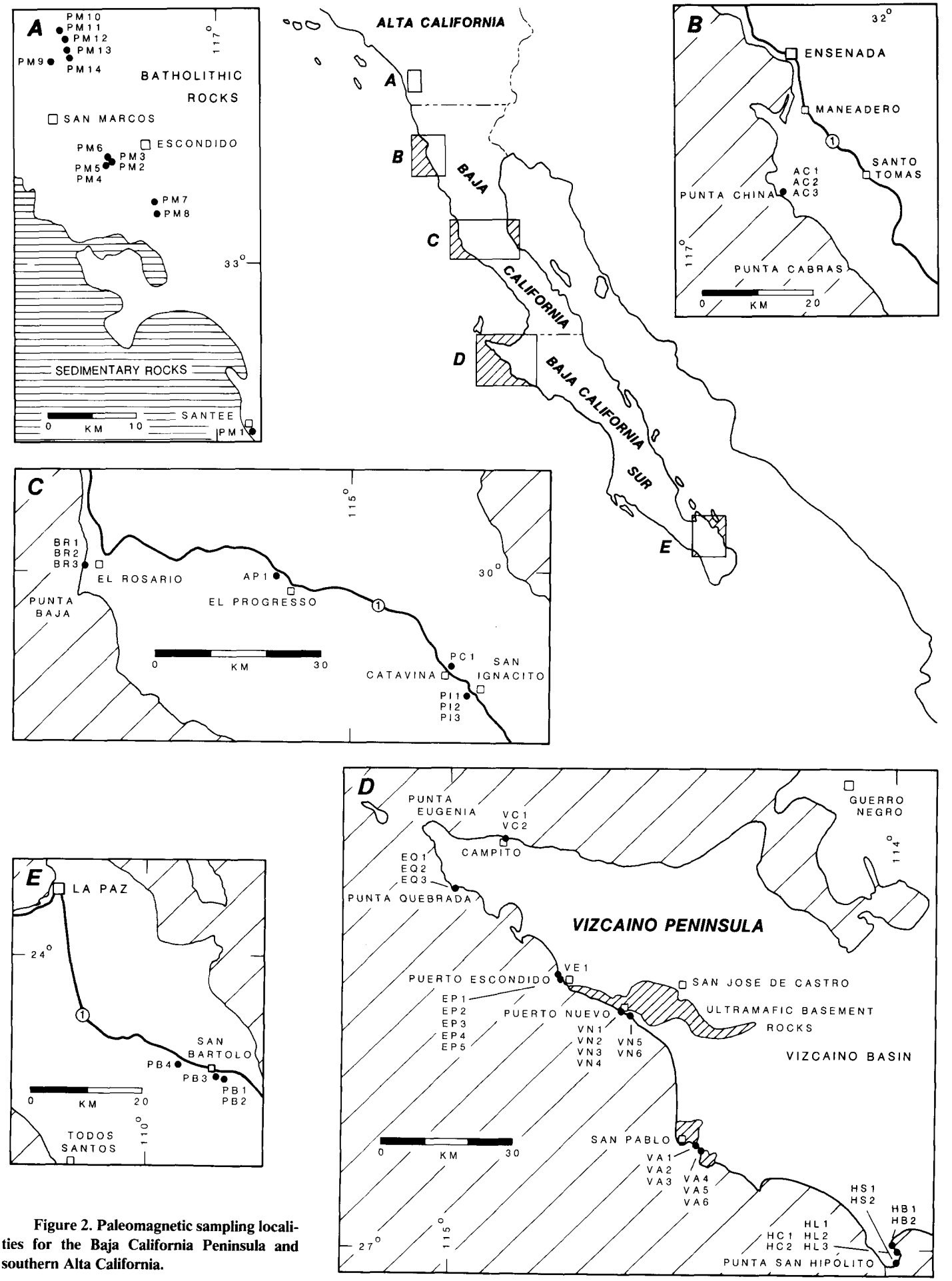


Figure 2. Paleomagnetic sampling localities for the Baja California Peninsula and southern Alta California.

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Cylindrical specimens 2.2 cm in length were cut from the field-drilled cores. All samples, except for the majority of those from the San Marcos Gabbro, were subjected to either progressive alternating-field (AF) or thermal demagnetization. AF demagnetization in peak fields between 1 and 100 mT was the primary technique used in this study. Stepwise thermal demagnetizations to peak temperatures between 600 and 680 °C were performed on one or two specimens from most sites and on samples from sites where AF demagnetization proved ineffective. In some instances, thermal demagnetization was continued on specimens which had already been subjected to AF treatment in order to isolate a high-coercivity component of magnetization remaining in the rock. Curie temperatures (T_c) were measured on 0.6-cm-diameter cores or magnetic separates from representative specimens with a recording magnetic balance that determined strong-field magnetization as a function of temperature (J_s-T). To aid further in defining the magnetic mineralogy, especially that of hematite mixed with titanomagnetite, progressive isothermal remanent magnetizations (IRM) were produced in selected cores using direct fields up to 0.7 T.

Components of magnetization were selected from demagnetization data using a least-squares method based on principal component analysis modified from Kirshvink (1980). Either lines (a single component) or planes (two components with overlapping coercivity, H_c, or blocking temperature, T_b, spectra) were fitted to the magnetization vector endpoints. Treatment of the samples from the San Marcos Gabbro differed, in that stepwise thermal and AF demagnetizations were performed on pilot samples from each site to determine the optimum cleaning level for the other samples. The statistical methods of Fisher (1953) are reported in the analysis of the site- and unit-mean averages, but Bingham statistics (Onstott, 1980) were used to analyze the Punta San Hipolito pillow basalts, which exhibit intersecting demagnetization planes. Only average inclinations (McFadden and Reid, 1982) were estimated for the Eugenia pillow basalts because of large between-site declination differences that are likely due to incomplete structural corrections (MacDonald, 1980).

PALEOMAGNETIC EXPERIMENTS

Punta San Hipolito Section

An Upper Triassic marine section at Punta San Hipolito constitutes the oldest strata on the Vizcaino Peninsula (Fig. 2D). The section is a 2,400-m-thick sequence of sedimentary rocks, the San Hipolito Formation, that conformably(?) overlies ophiolitic rocks consisting of altered low-potassium pillow basalts (Finch and others, 1979). Some interpillow zones are filled with red dolomitized limestone that contains the Middle to Late Triassic hemipelagic pelecypod *Halobia* sp. Chert, limestone, breccia, and sandstone members of the San Hipolito Formation are inferred to have been deposited oceanward of a volcanic arc that served as a source for these sedimentary rocks (Finch and Abbott, 1977). The chert member is predominantly celadonitic (green) with some hematitic (red) layers; it also contains *Halobia* sp. and a diverse assemblage of Late Triassic radiolarians (Pessagno and others, 1979). Flame structures within the chert show that the section is not overturned. The overlying limestone is thinly bedded, very finely recrystallized, and contains the hemipelagic pelecypod *Monotis cf. M. subcircularis* of Late Triassic (Norian) age near the top of the member (Finch and Abbott, 1977). The limestone also contains numerous zones with chaotic folds and autobreccias that were formed during soft-sediment slumping. The stratigraphically highest sandstone member consists of poorly sorted, calcite-, chlorite-, and zeolite-cemented volcaniclastic sediments that yield radiolarians indicating an Early Jurassic age (Whalen and Pessagno, 1984).

The Punta San Hipolito section was sampled for paleomagnetic study at nine sites (Fig. 2D). Each of two sites is in the pillow basalts (HB1 and HB2), chert (HC1 and HC2), and sandstone (HS1 and HS2) members, and three are in the limestone member (HL1–3). Only red hematitic chert beds were sampled, and samples from one site in the limestone member (HL2) are distributed around a single folded bed to test whether the magnetization predates soft-sediment deformation within the unit. NRM intensities (J_{NRM}) vary with lithology, and average values range from 5×10^{-3} Am^{-1} for the limestones, through 2×10^{-2} Am^{-1} for the cherts and 4×10^{-1} Am^{-1} for the sandstones, to 1 Am^{-1} for the pillow basalts. Upon AF and thermal demagnetization, all sites, except the cherts and pillow basalts, show a similar magnetization direction which is the only stable component of magnetization observed in most of the section. Random magnetizations of low H_c and T_b are usually removed by 10 mT or 100 °C treatment, and magnetization intensities are reduced to small values (<10% of J_{NRM}) by 50–70 mT or 550–565 °C. These results, together with a T_c determination in the sandstone member (535 °C) and IRM measurements in all members, indicate that titanomagnetite is the carrier of the stable remanence in all rocks except the cherts. Growth of a new magnetic phase in the limestone samples during laboratory heatings above 500 °C generally results in increased remanent intensities that reach several times the original NRM value by 650 °C.

A second component of magnetization is revealed by AF demagnetization in the pillow basalts and by thermal demagnetization in the red cherts. Overlapping coercive force spectra of the two components in the pillow basalts require the analysis of demagnetization planes (Halls, 1978) to determine the direction of the high-H_c component. Complete superposition of the two components in three of the nine stable samples from the basalts prevents the use of demagnetization planes, although these composite directions do lie between those of the two end-member components. Nearly complete demagnetization within 70 mT or 550 °C (Fig. 3A), IRM experiments, and a T_c of 536 °C indicate that titanomagnetite carries the stable magnetic remanence in the pillow basalts. Thermal demagnetization of chert samples removes the low-H_c component common to the other rock members between 100 and 560 °C (5–100 mT) and the high-H_c component between 550 and 670 °C (Fig. 3B). IRM experiments also show that the high-H_c component in the red cherts is associated with hematite. The high-H_c component of magnetization is characterized by positive inclinations in the chert member and negative inclinations in the pillow basalts.

Site-mean directions and their 95% confidence limits for the Punta San Hipolito section (Table 1) are shown before and after application of the structural correction in Figures 4A and 4B, respectively. Dispersion of the directions from the folded limestone bed (HL2) increases markedly after unfolding, with a significant decrease in the precision parameter (k) from 84 to 9 (significant at the 99% level; Watson and Irving, 1957), demonstrating that the limestone was folded before acquiring its magnetization. This component of magnetization, observed throughout the section (nine sites), is inferred to be an overprint, perhaps acquired by burial (Pullaiah and others, 1975) or by intrusion of these Triassic rocks by the Middle Jurassic to Lower Cretaceous tonalites (Moore, 1984). In contrast, the high-H_c and high-T_b components of the red cherts and pillow basalts (four sites) are considered to be earlier magnetizations because, although associated with different magnetic minerals and of different polarities, their absolute directions are consistent between these two units of similar age.

Eugenia Formation

Exposed primarily in fault-bounded blocks along the western margin of the Vizcaino Peninsula, the Eugenia Formation consists of immature

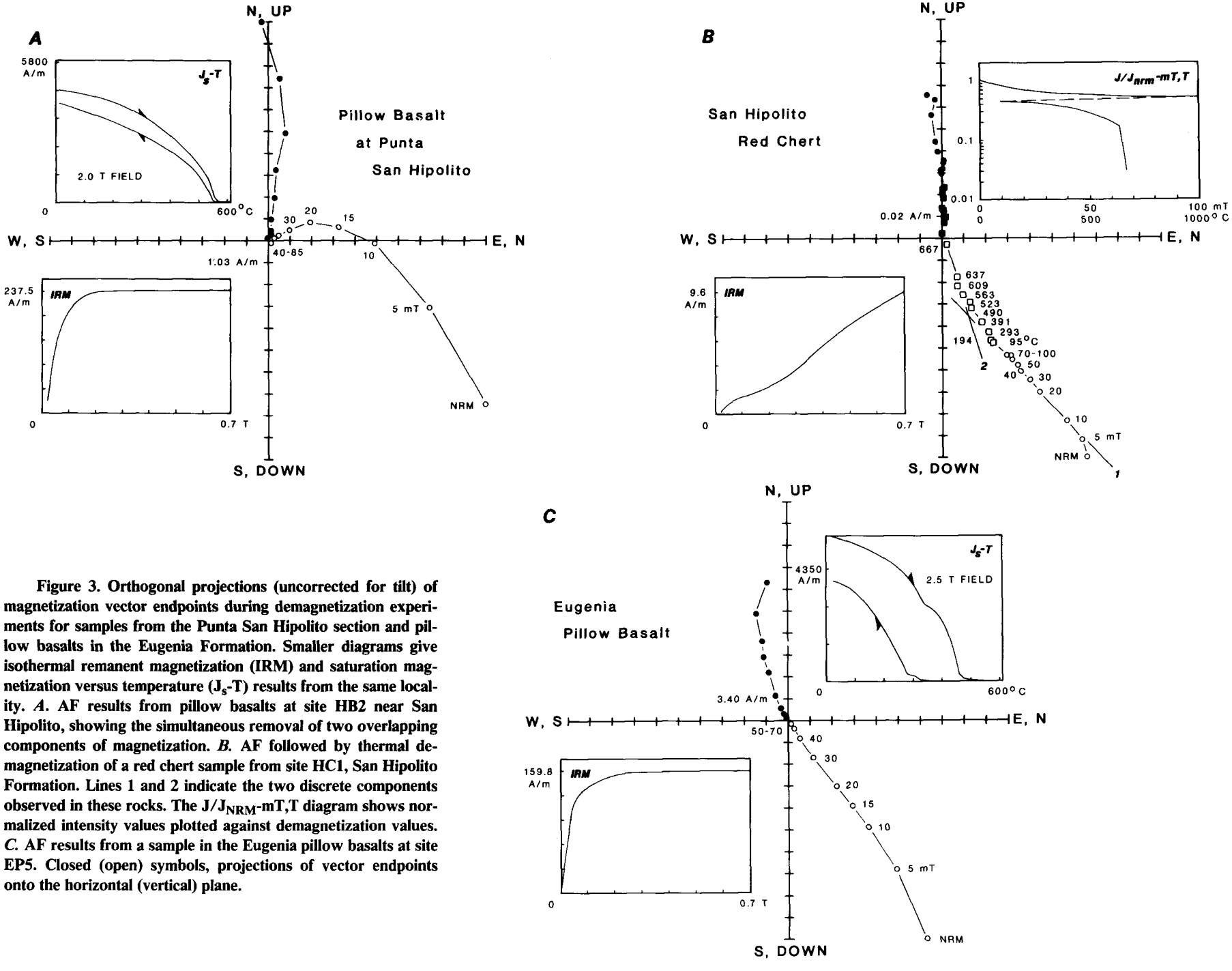


Figure 3. Orthogonal projections (uncorrected for tilt) of magnetization vector endpoints during demagnetization experiments for samples from the Punta San Hipolito section and pillow basalts in the Eugenia Formation. Smaller diagrams give isothermal remanent magnetization (IRM) and saturation magnetization versus temperature (J_s-T) results from the same locality. *A*. AF results from pillow basalts at site HB2 near San Hipolito, showing the simultaneous removal of two overlapping components of magnetization. *B*. AF followed by thermal demagnetization of a red chert sample from site HC1, San Hipolito Formation. Lines 1 and 2 indicate the two discrete components observed in these rocks. The J/J_{NRM}-mT,T diagram shows normalized intensity values plotted against demagnetization values. *C*. AF results from a sample in the Eugenia pillow basalts at site EP5. Closed (open) symbols, projections of vector endpoints onto the horizontal (vertical) plane.

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TABLE 1. SITE-MEAN PALEOMAGNETIC DIRECTIONS FROM PENINSULAR CALIFORNIA

Site	Uncorrected I	Uncorrected D	Corrected I	Corrected D	N	R	k	α_{95}	Site poles Plat	Site poles Plong
Punta San Hipolito Section (Upper Tr–Lower Jr)										Uncorr/(Corr)
HS1	63.0	350.5	71.7	254.0	5	4.8624	29	14.4	71.0	224.8
HS2	68.3	341.6	52.2	252.3	5	4.8745	32	13.8	61.9	221.3
HL1	47.0	355.3	46.3	303.3	5	4.9397	66	9.5	85.7	172.7
HL2	57.5	348.5			5	4.9525	84	8.4	75.2	207.9
HL2			62.3	259.2	5	4.5488	9	27.2	(11.1)	(199.5)
HL3	48.0	357.8	61.2	288.8	5	4.9848	264	4.7	87.2	202.1
HC1	43.8	348.0	51.8	261.2	5	4.8212	22	16.5	79.2	151.4
HC1′	80.5	29.3	35.5	210.2	5	4.9279	56	10.4	(34.9)	(30.8)
HC2	46.6	344.5	43.6	277.0	5	4.9825	228	5.1	76.2	163.1
HC2′	83.5	71.3	37.7	219.1	5	4.9223	51	10.8	(28.8)	(23.9)
HB1	47.5	334.7	39.9	259.8	5	4.8580	28	14.7	67.6	166.1
HB2	40.0	1.6	60.5	264.0	4	3.9915	354	4.9	85.5	46.6
HB12′	-73.0	146.8	-31.0	50.1	6	5.9494	99	12.5	(24.6)	(12.1)
Eugenia Formation (Upper Jurassic)										Corrected
EP1	47.8	324.3	33.5	350.0	9	8.6697	24	10.7	77.0	112.2
EP2	70.6	6.6	37.9	21.6	7	6.8411	38	10.0	69.4	348.2
EP3	79.4	40.8	45.5	34.3	8	7.8528	48	8.1	59.6	328.3
EP4	52.5	349.8	24.1	354.6	6	5.9008	50	9.5	74.2	85.1
EP5	63.9	340.2	33.1	336.3	6	5.8994	50	9.6	66.2	136.7
EQ1	44.1	345.3	8.9	358.9	6	5.5459	11	21.1	66.7	67.9
EQ2	48.1	329.0	18.5	351.4	6	5.5093	10	22.0	70.0	90.5
EQ3	31.1	336.3	-7.3	341.8	6	5.3367	8	26.1	11.0	226.5
Alisitos Formation (Lower Cretaceous)										Uncorrected
AC1	47.3	12.4	70.4	311.3	5	4.8724	31	13.9	78.9	346.1
AC2	48.0	3.5	65.1	303.7	5	4.9705	136	6.6	86.2	11.7
AC3	44.6	7.9	66.5	314.4	5	4.9664	119	7.1	81.3	8.5
AP1	56.9	16.9	2.8	33.7	5	4.8933	37	12.7	74.1	302.1
Peninsular Ranges Batholith (Cretaceous)										Uncorrected
PB1	45.2	17.6			5	4.9582	96	7.9	73.8	325.7
PB2*										
PB3	50.9	343.3			4	3.8739	24	19.2	73.3	192.2
PB4	54.5	20.3			5	4.9862	289	4.5	69.1	302.8
PI1	54.2	6.1			6	5.9867	377	3.5	82.8	289.2
PI2	56.9	8.9			5	4.9279	55	10.4	79.2	286.6
PI3	50.0	7.6			6	5.9764	212	4.6	83.3	323.5
PC1	54.6	358.1			5	4.7970	20	17.7	84.3	229.5
PM1	49.6	6.5			10	9.9374	144	4.0	84.0	354.4
PM2	44.2	4.4			10	9.9702	302	2.8	81.8	33.5
PM3	37.7	2.1			10	9.9602	226	3.2	77.9	53.6
PM4	53.1	3.2			10	9.9743	350	2.6	87.3	320.1
PM5	46.3	5.9			9	8.7514	32	9.2	82.5	18.4
PM6	45.2	357.8			10	9.7801	41	7.6	83.4	80.0
PM7	56.3	21.7			10	9.9380	145	4.0	71.8	314.6
PM8	55.6	13.8			10	9.6495	26	9.7	78.2	313.7
PM9	45.9	2.6			11	10.9537	216	3.1	83.7	41.0
PM10	45.0	15.8			9	8.9765	340	2.8	74.8	354.4
PM11	48.9	18.1			10	9.9779	408	2.4	74.2	340.3
PM12	48.0	11.3			10	9.9863	655	1.9	79.5	353.5
PM13	46.6	18.4			10	9.9903	927	1.6	73.3	346.5
PM14	43.8	11.0			10	9.9786	420	2.4	77.8	8.5
Valle Formation (Upper Cretaceous)										Uncorr/(Corr)
VE1	45.0	53.4	29.1	18.0	5	4.9372	64	9.7	(69.4)	(7.3)
VN1	37.0	9.0	46.8	53.5	7	6.9291	85	6.6	79.3	13.5
VN2	34.1	0.7	51.5	44.9	5	4.9795	195	5.5	81.2	60.8
VN3	38.0	0.3	53.5	50.3	5	4.9704	135	6.6	83.8	62.9
VN4	40.0	6.1	50.2	56.2	5	4.9650	114	7.2	82.7	14.9
VN5	43.2	31.4	58.1	65.0	3	2.9749	80	13.9	61.9	333.0
VN6	38.2	34.2	61.5	58.5	6	5.9351	77	7.7	58.4	338.7
VC1	66.1	329.2	48.4	3.3	6	5.9914	585	2.8	(86.7)	(305.7)
VC2	64.9	338.6	45.1	6.8	6	5.9698	166	5.2	(83.8)	(344.6)
VA1	32.1	1.7	60.7	328.8	6	5.9397	83	7.4	80.0	56.5
VA2	40.4	8.5	70.7	320.1	6	5.5259	68	8.2	81.3	2.4
VA3	40.6	13.8	65.5	317.6	7	6.9175	73	7.1	76.9	350.5
VA4	45.8	19.9	52.6	61.4	8	7.9357	109	5.3	72.3	331.2
VA5	43.8	14.0	54.8	54.4	5	4.9674	123	6.9	77.4	339.9
VA6	42.6	4.8	59.6	44.9	7	6.9334	90	6.4	85.0	5.6
La Bocana Roja Formation (Upper Cretaceous)										Corrected
BR1	56.8	356.0	51.2	9.1	6	5.9939	820	2.3	82.0	318.7
BR2	57.5	5.0	50.6	16.9	6	5.9874	413	3.3	75.4	325.0
BR3	59.9	355.0	54.2	9.7	5	4.9921	507	3.4	80.5	301.8

Note: uncorrected, field directions or poles uncorrected for tilt; Corrected, directions or poles corrected for tilt; I, inclination in degrees; D, declination in degrees; N, number of samples averaged; R, vector sum; k, precision parameter; α_{95}, radius of 95% confidence in degrees; Plat, Plong, latitude and east longitude of virtual geomagnetic poles in degrees; ′, components of higher coercivity or blocking temperatures observed at the same site or sites (see text); *, discarded due to unstable demagnetization behavior; HB12′, direction resolved with demagnetization planes and Bingham statistics (Onstott, 1980) from samples of sites HB1 and HB2, α_{95} is an average of the two Bingham 95% confidence limits, $k = k_1$ of the two Bingham concentration parameters, and R is calculated from k_1.

marine-fan sediments and intercalated volcanic rocks that include tuffs, flows, and pillow lavas (Barnes and Berry, 1979; Boles and Hickey, 1979). A composite section totals ~2,300 m in thickness and contains the mollusk *Buchia piochii* and radiolarian assemblages of Late Jurassic (Portlandian) age (Minch and others, 1976; Boles, 1978). Conglomerate horizons in the lower part of the formation indicate significant contemporaneous tectonic activity in an adjacent volcanic arc (Minch and others, 1976). The Lower Cretaceous uppermost part consists predominantly of well-sorted, fine-grained sandstones that contain abundant biotite and quartz, indicating a closer proximity to cratonal sources in the east (Boles and Hickey, 1979; Moore, 1983).

Pillow basalts of the Eugenia Formation were collected at five sites in

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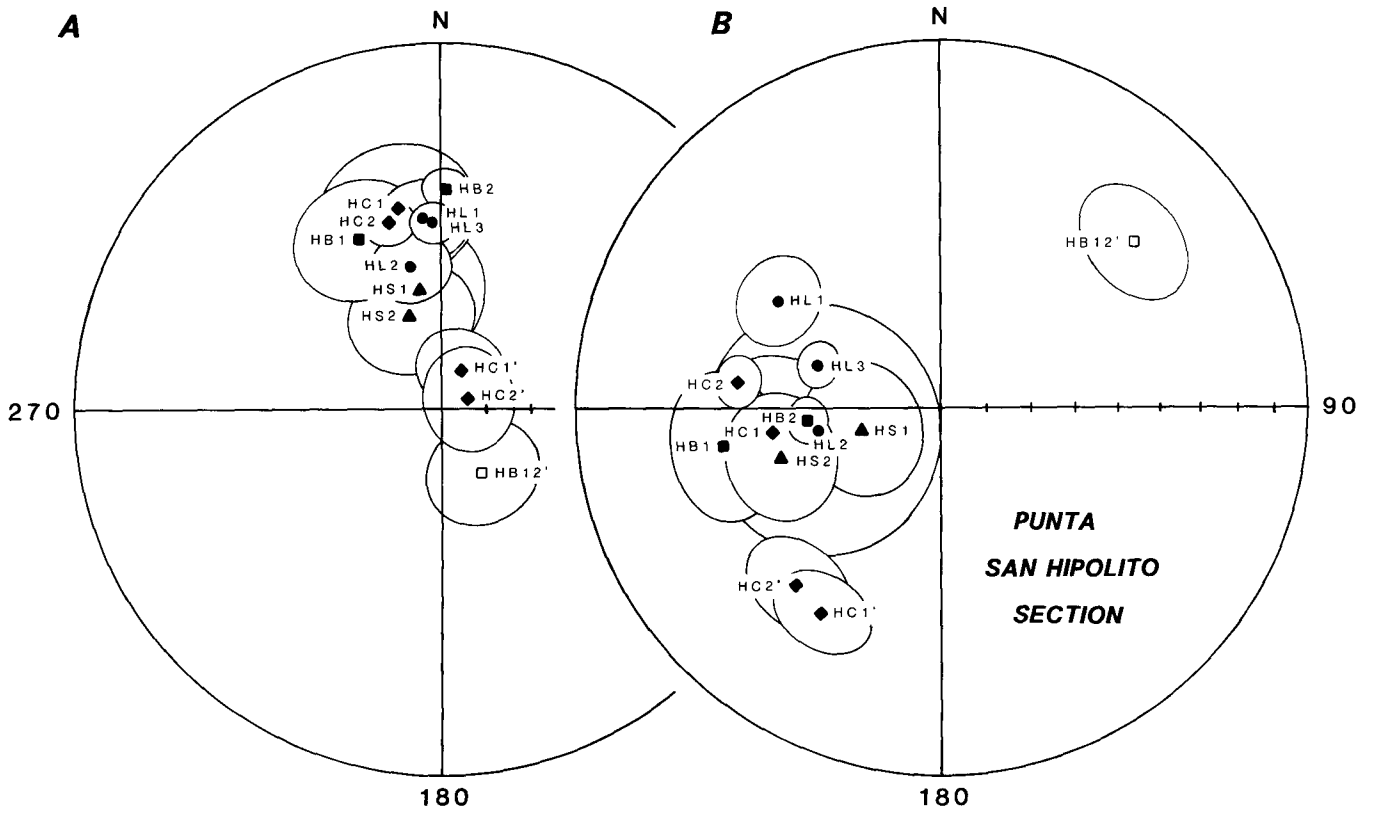


Figure 4. Site-mean directions and 95% confidence limits from the Punta San Hipolito section. *A* shows directions uncorrected for bedding tilt and *B* shows corrected directions. Diagrams are equal-area plots in which open symbols indicate projection from the upper hemisphere, and closed symbols, from the lower hemisphere. Large inverted triangle symbol indicates present-day field direction.

the unweathered beach-cliff exposures near Puerto Escondido and from three sites in similar exposures at Punta Quebrada (Fig. 2D). To provide better structural attitude control, sites were sought near interbedded sedimentary horizons. A single component of magnetization was removed from the pillow basalts during demagnetization, usually between 10 and 50 mT or 250 and 560 °C (Fig. 3C). Curie temperatures ranging from 446 to 454 °C and IRM measurements support the demagnetization results in demonstrating that titanomagnetite is the dominant magnetic mineral in these rocks.

The site-mean directions (Table 1) plotted in Figure 5 are dispersed in declination after structural correction, particularly for sites EP2 and EP3. The variation is most likely due to incomplete restoration of structural attitudes, as numerous high-angle faults cut the sampled exposures and perhaps introduce unmeasurable vertical-axis rotations between sites. A unit-mean inclination of 24.4 ± 15.1 (corrected for structural tilt) is calculated from only the inclination values (McFadden and Reid, 1982) to circumvent these possible declination errors. Although no significant improvement in dispersion of the inclination values is observed upon tectonic correction (due primarily to the similarity in dips measured at the sampling sites), the high-T_b remanent magnetization is considered a TRM acquired during cooling of these relatively unaltered pillow basalts (for example, Barnes and Berry, 1979).

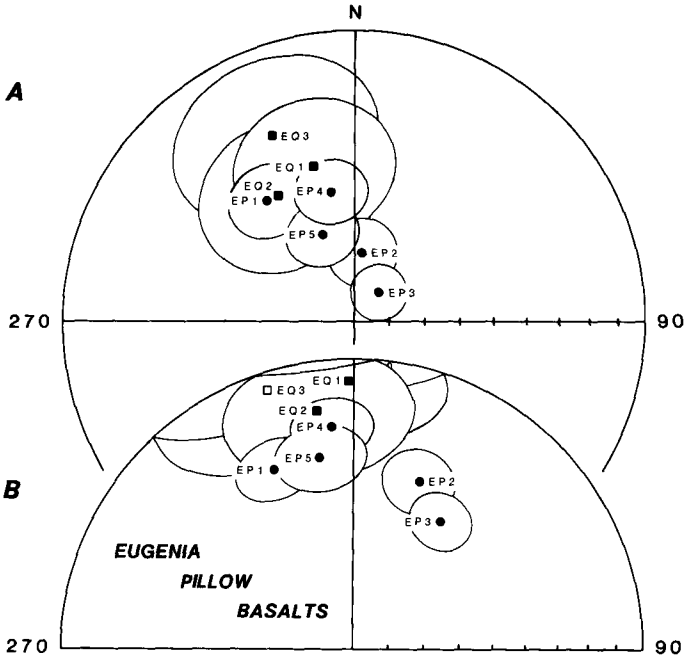


Figure 5. Site-mean directions and 95% confidence limits for the pillow basalts in the Eugenia Formation from Punta Quebrada (EQ1 and EQ2) and near Puerto Escondido (EP1–5). *A*. uncorrected for stratal tilt. *B*. corrected for tilt.

Alisitos Formation

The Alisitos Formation of the northern state of Baja California consists of at least 3 km of flow and pyroclastic volcanic rocks, volcaniclastic

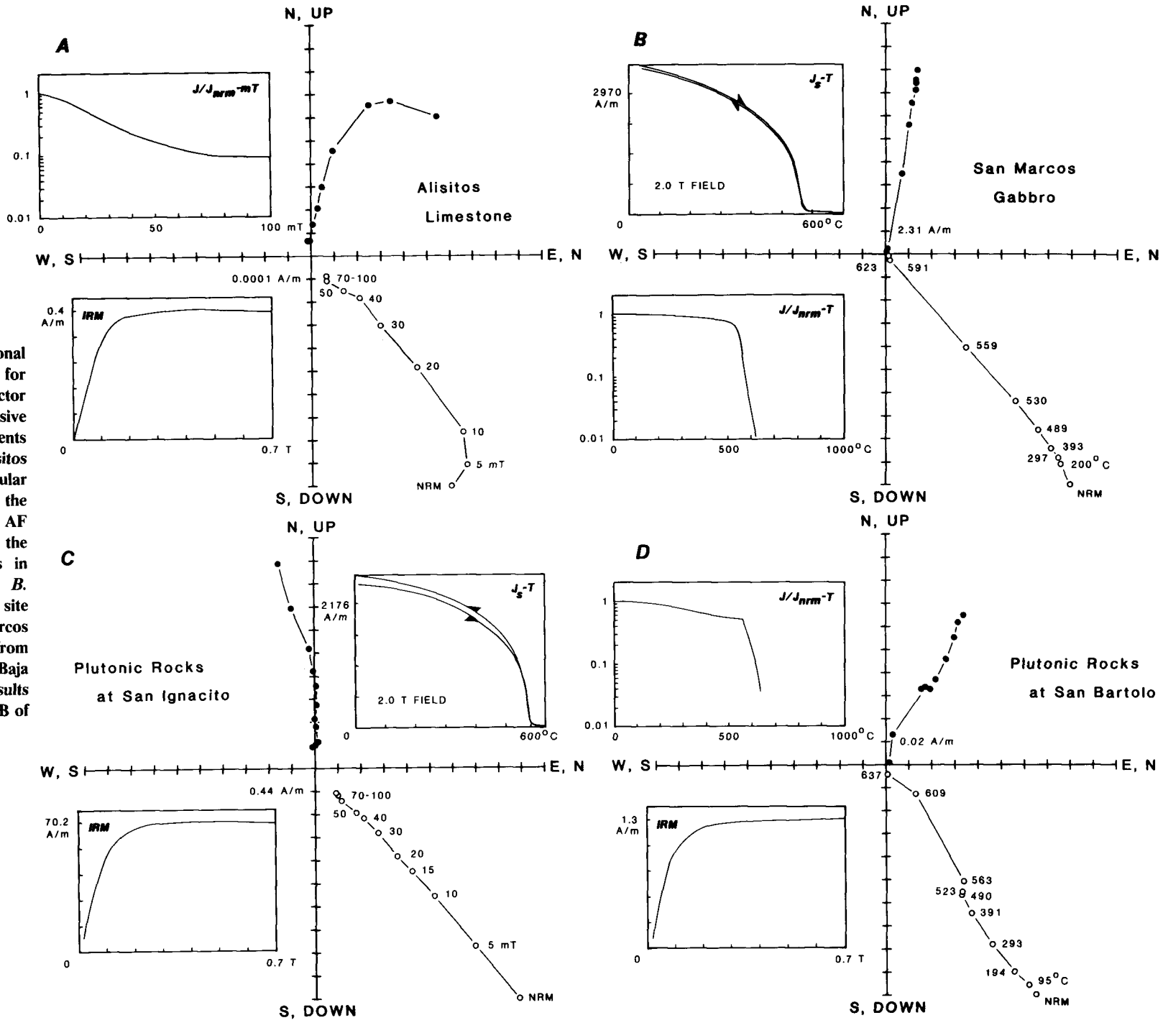


Figure 6. Orthogonal projections (uncorrected for tilt) of magnetization vector endpoints during progressive demagnetization experiments for samples from the Alisitos Formation, the Peninsular Ranges batholith, and the San Marcos Gabbro. *A*. AF results from site AC2 in the remagnetized limestones in the Alisitos Formation. *B*. thermal results from site PM12 in the San Marcos Gabbro. *C*. AF results from site PI3 in the PRB of Baja California. *D*. thermal results from site PB4 in the PRB of Baja California Sur.

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sedimentary rocks, and sparse but locally massive accumulations of biohermal limestones (Silver and others, 1963; Beggs, 1984). A middle Albian age is assigned to the caprinid limestone at Punta China (Allison, 1955; Fig. 2B). Coastal sections of the Alisitos Formation have undergone low-grade metamorphism (Silver and others, 1963), and the degree of metamorphism and alteration within the formation increases eastward toward the plutonic rocks of the Peninsular Ranges. Near Punta Cabras (Fig. 2B), the Alisitos Formation is intruded by shallow diorite to granodiorite plutons that yield a U-Pb age of 115 ± 2 m.y. (Silver and others, 1969). Structural deformation of the Alisitos Formation in this region formed north-northwest–trending folds and occurred within an approximate 10-m.y. interval coincident with the plutonism and related volcanism (Silver and others, 1969).

Oriented samples were collected from biohermal limestones of the Alisitos Formation in an active quarry at Punta China (AC1–3) and inland from a roadside outcrop near El Progresso (AP1, Fig. 2C). Large fossil masses were avoided so that only samples of the relatively homogenous interstitial limestone were obtained. NRM intensities were low ($J_{NRM} = 3 \times 10^{-4}$ Am^{-1}), and a single component of magnetization was observed in the limestones, generally between 30 and 70 mT or 100 and 450 °C (Fig. 6A). As with the limestone member of the San Hipolito Formation, heating above 400–500 °C created a new magnetic phase with high thermoremanent magnetization (TRM) capacity. IRM experiments show that magnetite in these rocks is accompanied by a low-intensity, high-H_c mineral.

A statistically significant grouping (99% confidence level) of the site-mean directions before application of the structural corrections, rather than afterward, indicates that the Alisitos limestones were magnetized after folding or tilting of the units (Figs. 7A and 7B). The remagnetization of the limestones is most likely associated with the alteration of the unit caused by nearby plutons of the Peninsular Ranges batholith. A comparison of the uncorrected mean directions from the Alisitos Formation with those from the batholithic rocks (next section) shows that they are not significantly different at the 95% confidence level.

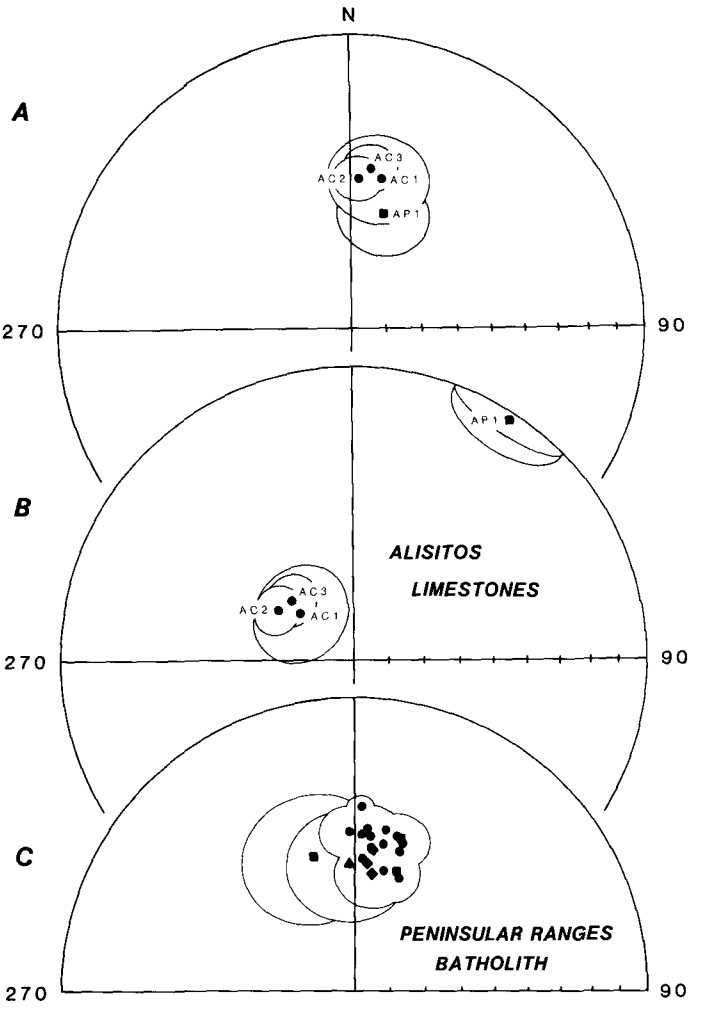
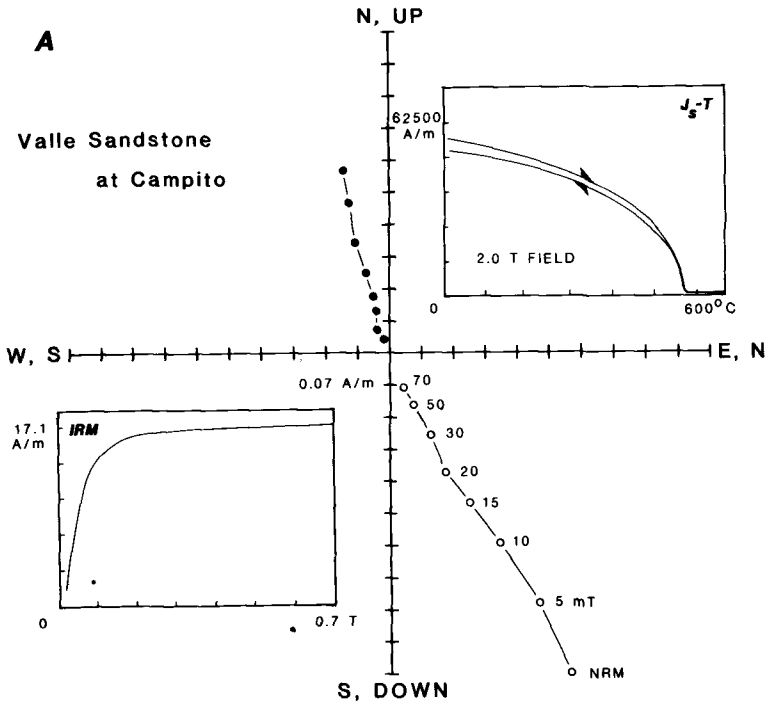


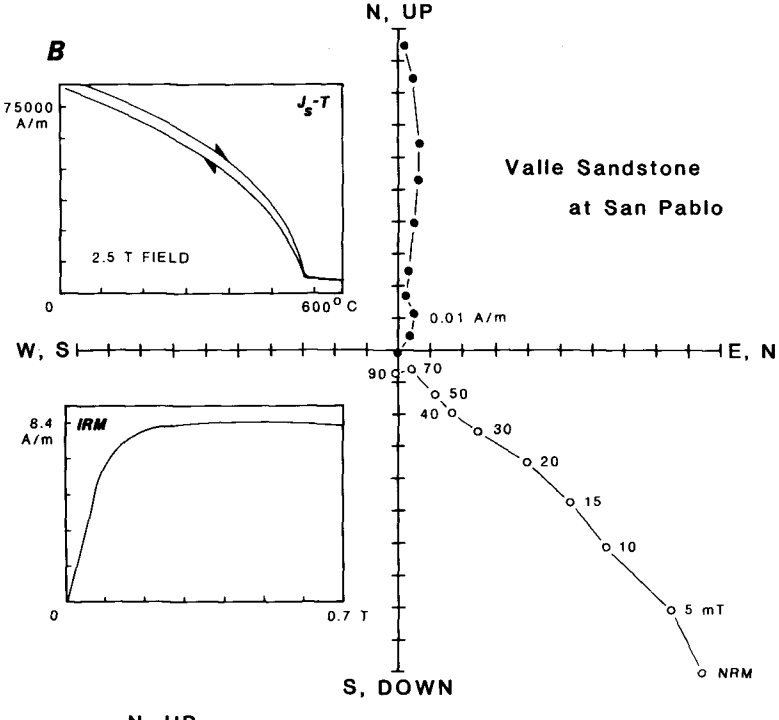
Figure 7. Site-mean directions from *A*, limestones in the Alisitos Formation at Punta China (AC1–3) and near El Progresso (AP1), uncorrected for tilt; *B*, corrected for tilt; and *C*, the Peninsular Ranges batholith and San Marcos Gabbro. In *C*, square symbols indicate mean directions for localities near San Bartolo; triangle, near Catavina; diamonds, near San Ignacito; and circles, near San Marcos.

Peninsular Ranges Batholith

Hundreds of plutons constitute the Peninsular Ranges batholith. They range in diameter from 1 to >50 km (Gastil and others, 1975). Most are composed of tonalite and granodiorite, although compositions range from syenogranite to gabbro (Gastil and others, 1975). Consistent isotopic and stratigraphic data yield emplacement ages for the plutons ranging from 90 to 120 m.y. (Silver and others, 1979), although potassium-argon ages are usually somewhat younger and reflect the cooling history of the plutonic rocks (Krummenacher and others, 1975). The K-Ar age for the San Marcos Gabbro of southern Alta California (Fig. 2A) is ~108 to 113 m.y. (Dalrymple, 1976; all K-Ar ages converted to new constants after Dalrymple, 1979). For the axial region of the Peninsular Ranges batholith in the northern Baja Peninsula, K-Ar age determinations generally range between 77 and 87 m.y. (Gastil and others, 1975), and from the Cabo San Lucas area at the southern tip of the peninsula (Fig. 1), granitoids range in age between 72 and 113 m.y. (Gastil and others, 1976). Two of the southernmost localities, also sampled for paleomagnetic analysis, give K-Ar biotite ages of 72.9 ± 2.2 and 92.7 ± 2.8 m.y. (V. A. Frizzell, Jr., and K. M. Ort, 1983, written commun.).

In the San Marcos Gabbro of southern California (Fig. 2A), 14 sites in quarries and roadcuts were sampled. These sites correspond, in part, to some of those sampled by Teissere and Beck (1973). A mafic phase within a granodiorite body, as well as in crosscutting aplite and diabase dikes, were sampled at three stream-wash sites near San Ignacito (PI1–3), and a massive, medium-grained granodiorite pluton was sampled near Catavina (PC1) in the state of Baja California (Fig. 2C). To the south, four sites were sampled from fine-grained quartz-monzonitic rocks (PB1–4) near San Bartolo, Baja California Sur (Fig. 2E). In these samples, higher average NRM intensities are associated with the more mafic rocks sampled at San Marcos (4 Am^{-1}) and San Ignacito (5×10^{-1} Am^{-1}), intermediate values (2×10^{-2} Am^{-1}) are associated with those near San Bartolo and the aplite dike at San Ignacito; and the lowest values (10^{-3} Am^{-1}), with the pluton at Catavina. Generally, a single component of magnetization was observed in these rocks during demagnetization. This component is removed between 10 and 100 mT or, primarily, between 500 and 590 °C in the San Marcos (Fig. 6B) and San Ignacito rocks (Fig. 6C); however, ~40% to 50% of the NRM intensity remains in the Catavina and San Bartolo rocks after AF treatment to 100 mT (Fig. 6D). Heating to 580 °C reduces the intensities of the San Bartolo samples to <10% of the NRM intensities; intensities of the Catavina samples decay to ~20% of NRM intensity by 500 °C. Increases in intensity due to a mineralogy change are seen in the Catavina samples after heating above 500 °C. IRM results and T_c values ranging from 553 to 582 °C show that titanomagnetite is the dominant magnetic mineral in these plutonic rocks. Samples from one site near San Bartolo give erratic behavior upon demagnetization and have





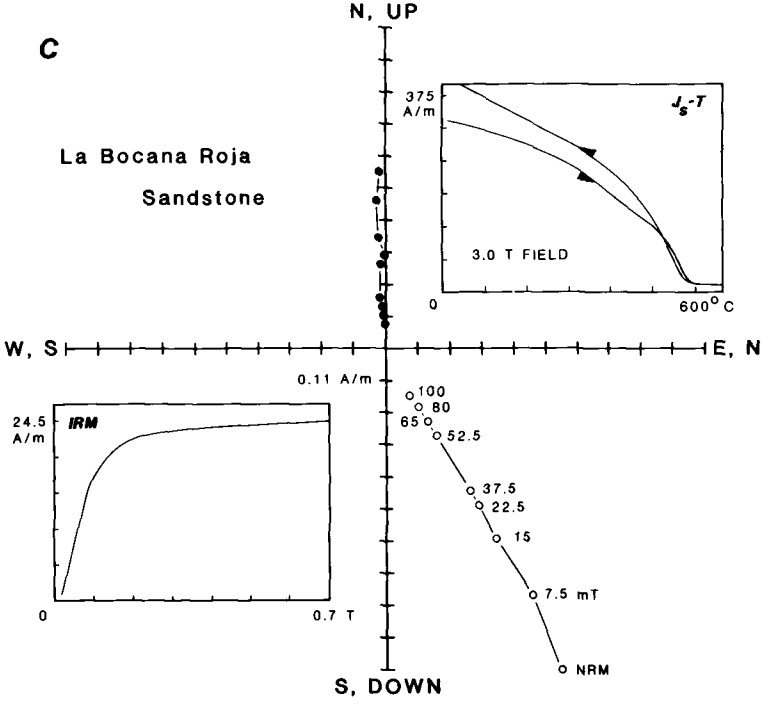


Figure 8. Orthogonal projections (uncorrected for tilt) of magnetization vector endpoints during progressive AF demagnetization experiments: *A*, from site VC2 in the Valle Formation at Campito; *B*, from site VA2 in the remagnetized Valle at San Pablo; and *C*, from site BR1 in the Bocana Roja Formation near El Rosario (Fig. 2).

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been discarded from further consideration. The stable high-T_b component of magnetization resolved in these plutonic rocks is inferred to be a primary TRM.

The site-mean directions from the plutonic rocks of peninsular California are plotted in Figure 7C. There is no significant difference between the paleomagnetic poles from plutons in southern California, the northern Baja California Peninsula, and those from the southernmost portion of the peninsula.

Valle Formation

Widely distributed across the western Vizcaino Peninsula, the Valle Formation is a thick (10-km) turbidite sequence that represents a deep-sea fan complex deposited along the margin of an oceanic basin at bathyal depths (Minch and others, 1976). Diverse foraminiferal and calcareous nannoplanktonic faunules indicate an Albian to Santonian age for the formation (Berry and Miller, 1984). Postemplacement uplift, cooling, and unroofing of the Peninsular Ranges batholith provided the major source of sedimentary debris shed westward into a subsiding forearc basin (Patterson, 1979). Postdepositional thrusting and uplift have placed the lower part of the Valle Formation against serpentinized ultramafic basement rocks in the Sierra San Andres near Puerto Nuevo and farther south near San Pablo (Fig. 2D). Metasomatic alteration and mineralization associated with these serpentinites have affected the Valle Formation in the vicinity of these contacts (Minch and James, 1975).

The Valle Formation was sampled at two sites near Campito (VC1 and VC2), one site near Puerto Escondido (VE1), six sites near Puerto Nuevo (VN1–6), and six sites near San Pablo (VA1–6; Fig. 2D). Samples ranging from fine-grained sandstones to shale were collected, and site-average NRM intensities fall between 7×10^{-3} Am^{-1} and 5×10^{-2} Am^{-1}. At all sites, a single stable component of magnetization was characterized by coercivities between 10 and 70 mT or blocking temperatures between 100 and 400 °C (Figs. 8A and 8B). Directions of magnetization are erratic and associated with large increases in intensity after thermal treatment above 400 °C. The path of the magnetization vector endpoints is generally toward the origin, but at several sites, it converges to a point away from the origin, indicating the presence of an inconsistent high-H_c component. IRM acquisition experiments also show that magnetite (T_c = 573 to 578 °C) is accompanied by a higher-H_c mineral in the Valle Formation.

Although uniform in demagnetization behavior, the directions of magnetization from the various sampling localities exhibit some differences. Figures 9A and 9B show, respectively, the site-mean directions for the Valle Formation before and after correction for structural tilting. The directions for the sites at Campito (VC1 and VC2) and that near Puerto Escondido (VE1) converge upon a common direction after correction, suggesting that they were consistently magnetized before being tilted, whereas the sites near San Pablo (VA1–6) diverge after correction, suggesting that they were magnetized after being tilted. The group of site-mean directions from Puerto Nuevo (VN1–4) follow a subset of the sites near San Pablo (VA4–6) in moving from northerly declinations and moderate inclinations before correction, to east-northeast declinations and steeper inclinations after correction. The sampling sites at Puerto Nuevo and San Pablo, near exposures of ultramafic basement rocks (Fig. 2D), thus appear to have acquired their magnetization after tilting and may be remagnetized. Although these sites are distant from visible intrusive contacts, deformed and altered rocks (calcified) may have been preferentially but unwittingly selected as the freshest-appearing exposures with varying structural attitudes.

Petrographic observations confirm that the samples collected at Puerto Nuevo and San Pablo are more altered than those collected at Campito and Puerto Nuevo. Some opaque phases from the altered group occur along cleavages and fractures of other mineral grains, suggesting secondary origins, whereas those from the relatively unaltered group appear to be only detrital (J. G. Liou, 1984, personal commun.). Flat-lying Miocene sediments overlie both the Valle Formation and ultramafic rocks east of Puerto Nuevo and indicate that the deformation and, perhaps, alteration predate these rocks. A comparison of the directions from the corrected and uncorrected sites in the Valle Formation, together with the mean direction from Patterson (1984), shows that they are not significantly different at the 95% confidence level and suggests that the alteration and remagnetization occurred soon after the Valle Formation was initially magnetized. Uncorrected directions from the two Puerto Nuevo sites with differing structural attitudes (VN5 and VN6) conform with directions from the other sites from this locality after structural correction. This implies that a second deformational event affected these two sites after all of the rocks at Puerto Nuevo were remagnetized, possibly during Miocene tectonism (Robinson, 1979).

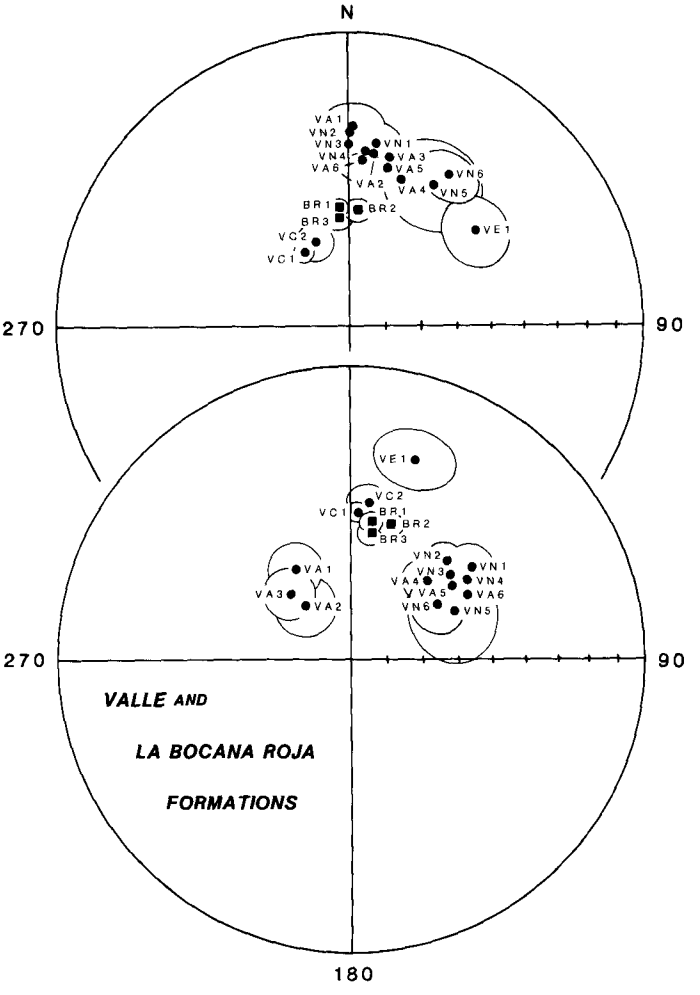


Figure 9. Site-mean directions from the Valle Formation at Campito (VC1 and VC2), near Puerto Escondido (VE1), at Puerto Nuevo (VN1–6), near San Pablo (VA1–6), and from the Bocana Roja Formation near El Rosario. *A.* uncorrected for tilt. *B.* corrected for tilt.

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La Bocana Roja Formation

The nonmarine La Bocana Roja Formation of Turonian(?) age is the oldest of four Upper Cretaceous formations recognized near El Rosario by Kilmer (1965). The sequence consists of marine and nonmarine sediments ranging from conglomerate to shale, including some interbedded tuffaceous rocks.

Fine-grained, reddish brown sandstones were sampled from a homoclinal beach-cliff exposure at three sites (BR1–3; Fig. 2C) near El Rosario. AF and thermal demagnetization results from these samples show a single consistent component of magnetization. AF treatment to 100 mT usually reduces the intensity of the samples to within 20% of their initial NRM intensity ($J_{NRM} = 8 \times 10^{-2}$ Am^{-1}; Fig. 8C), and demagnetization is virtually complete by heating to 620 °C. IRM results indicate that magnetite (T_c = 588 °C) and some hematite are present in these rocks. The uncorrected site-mean directions calculated for the Bocana Roja Formation (BR1–3) are plotted together with the directions from the similarly aged Valle Formation in Figure 8A. After correction for tectonic tilt, the Bocana Roja directions (Fig. 8B) are not significantly different from those of the Valle Formation at Campito, suggesting that both of these units acquired their magnetization under similar geomagnetic field conditions before being tilted.

TECTONIC IMPLICATIONS

The Cretaceous paleomagnetic poles of this study, together with those from previous studies (Teissere and Beck, 1973; Patterson, 1984), are plotted in Figure 10, showing that on an axial geocentric dipole hypothesis, the Baja California Peninsula most likely has been displaced farther northward (d = 11°; Table 2) and rotated farther clockwise (r = 32°) with respect to stable North America since the Cretaceous than can be accounted for by simply closing the Gulf of California (d = 2°, r = 8°). The paleomagnetic results also indicate that the Upper Triassic rocks of the Punta San Hipolito section were magnetized at a paleolatitude of 19.4 ± 11.6° and that the Upper Jurassic pillow basalts of the Eugenia Formation appear to have been magnetized at a paleolatitude of 13.5 ± 8.6° (Table 2). Because the precise ages and thus expected polarities for these Triassic and Jurassic rocks are unknown, the hemisphere (North or South) in which they formed is also uncertain. The uniformly normal polarities exhibited by the Cretaceous rocks of this study are, however, primarily due to the magnetization of these rocks during the Cretaceous Long Normal Polarity Interval (for example, Lowrie and Alvarez, 1981).

Alternate explanations to paleomagnetically inferred tectonic translations usually involve shallow-inclination errors for sedimentary rocks or unrecognized tilts for plutonic rocks (for example, Verosub, 1977; Beck, 1980). The consistent results between the Cretaceous sedimentary and plutonic rocks of this study over a wide, tectonically variable area apparently show that neither of these errors has had a significant effect. Other sources of error could arise from undetected remagnetizations or irregularities in the geomagnetic field. Widespread remagnetization under present-day field conditions appears unlikely, and the similarity between some of the discordant Cretaceous magnetization directions and the recent geomagnetic field direction is presumed coincidental. Local remagnetizations of sedimentary strata which succeed structural deformations have been detected by negative 'fold' tests in this study and, in most cases, yield uncorrected directions appropriate for the inferred age of the remagnitization. The extensive sampling of slowly cooled plutonic rocks and of thick sequences of sedimentary rocks also should eliminate temporal averaging problems related to shorter-period perturbations of the geomagnetic field (Merrill and McElhinny, 1983).

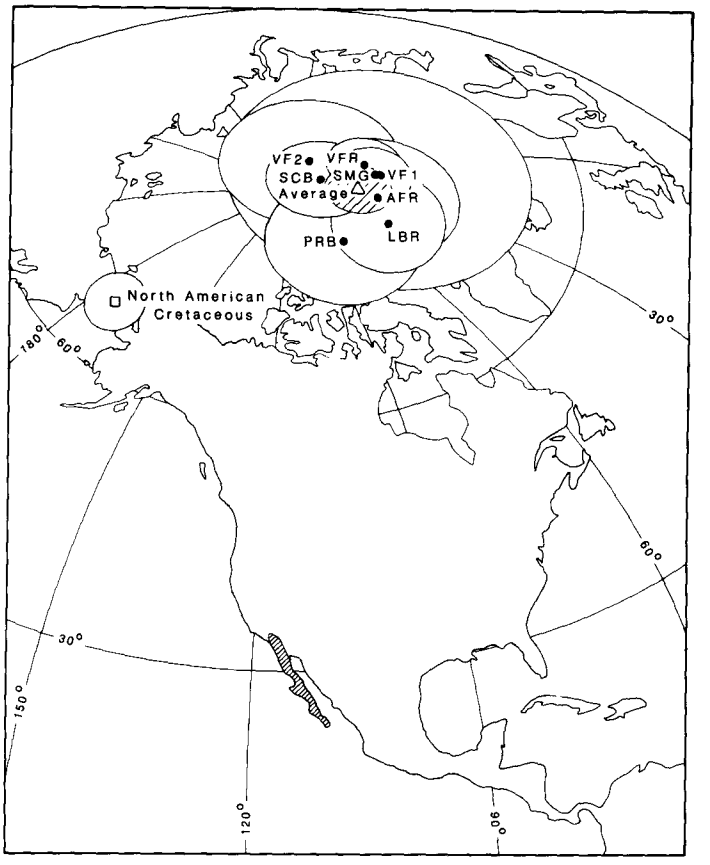


Figure 10. Equal-area map showing the Cretaceous mean paleomagnetic poles and reference paleomagnetic pole for cratonic North America (square) with 95% confidence limits (Table 2). Virtual geomagnetic poles are for the Peninsular Ranges batholith of Baja California and Baja California Sur (PRB), the Southern California batholith (SCB; Teissere and Beck, 1973), the San Marcos Gabbro (SMG), the Valle Formation (VF1, this study; VF2, Patterson, 1984; VFR, this study, remagnetized), the Bocana Roja Formation (LBR), and the remagnetized Alisitos Formation (AFR). The triangle symbol indicates the average value for the peninsular California rocks (shaded area).

In a study of the truncated continental margin along the northern Middle America Trench, Karig and others (1978) suggested that a portion of the margin had been tectonically removed by right-lateral transport sometime between Late Cretaceous and late Miocene time. Approximately 1,000 km of northward offset is needed to completely remove the margin south of Cabo Corrientes at the mouth of the Gulf of California (Fig. 1). On the basis of paleomagnetic results from the southern California batholith (Teissere and Beck, 1973), Beck and Plumley (1979) proposed that these rocks and, by inference, the entire Baja Peninsula, had initially formed adjacent to the now-truncated Middle America continental margin. The paleomagnetic results from this study support such a reconstruction. A restoration of the predicted 11° of northward movement and 32° of clockwise rotation places the Baja Peninsula against the northern portion of the truncated coastline (Fig. 11). A more southerly position (within the 95% confidence limits) is preferred, however; as this would place the peninsula against a greater length of the truncated margin and better align the contemporaneous arc segments. This reconstruction would

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TABLE 2. MEAN AND REFERENCE POLE POSITIONS, POLEWARD DISPLACEMENTS AND ROTATIONS BETWEEN THEM, AND OBSERVED AND EXPECTED PALEOLATITUDES

Rock unit	Plat	Plong	N	R	K	A_{95}	d	r	Paleolat	Ref
San Hipolito section	29.7	22.0	3	2.9712	70	14.9			19.4 ± 11.6	1
N. America (Triassic)	62.1	91.5	8	7.9337	106	5.4			1.5 ± 4.2	2
Eugenia Formation	..	..	..	..	..	..			13.5 ± 8.6	1
N. America (Jurassic)	70.4	109.2	10	9.8089	47	7.1			12.9 ± 5.5	3
Penin. Ranges batholith	81.7	285.5	7	6.8875	53	8.3	2.2 ± 7.0	30.7 ± 8.4	33.1 ± 6.5	1
San Marcos Gabbro	81.1	357.1	14	13.8837	112	3.8	13.9 ± 4.0	35.4 ± 5.0	29.1 ± 3.0	1
So. Calif. batholith	86.5	11.7	18	17.5947	42	5.4	12.0 ± 5.0	29.3 ± 6.1	33.1 ± 6.5	4
Valle Formation	80.6	356.4	3	2.9689	64	15.5	13.2 ± 12.4	34.6 ± 13.7	23.9 ± 12.1	1
Valle Formation	85.1	45.5	6	5.9087	55	9.1	14.1 ± 7.6	30.9 ± 8.4	22.9 ± 7.1	5
La Bocana Roja Fm.	79.5	316.6	3	2.9948	387	6.3	7.0 ± 5.6	37.1 ± 6.8	32.7 ± 4.9	1
Remagnetized units:										
San Hipolito section*	79.0	192.4	9	8.8264	46	7.7	3.2 ± 6.6	14.2 ± 7.9	33.2 ± 6.0	1
Alisitos Formation	81.3	336.1	4	3.9746	118	8.5	10.4 ± 7.1	35.2 ± 8.4	30.0 ± 6.6	1
Valle Formation	81.5	8.1	10	9.9383	146	4.0	14.3 ± 4.1	32.1 ± 4.7	22.7 ± 3.1	1
Baja Peninsula average	83.3	344.4	8	7.9734	263	3.4	10.9 ± 3.8	32.0 ± 4.5	26.7 ± 2.7	1
N. America (Cretaceous)	67.3	186.3	5	4.9921	505	3.4			37.6 ± 2.7	6

Notes as in Table 1. d, r, poleward displacement and clockwise rotation with respect to stable N. America, in degrees, with 95% confidence limits after Beck (1980), as modified by Demarest (1983); Paleolat, paleolatitude in degrees N for an average sampling location, 95% confidence limits as before, with value for the Eugenia Formation calculated from inclination data only, after McFadden and Reid (1982); * not included in Baja Peninsula average. References: 1, this study; 2, Irving and Irving (1982), Table 1, entries 56–63; 3, Irving and Irving (1982), entries 56–63; 4, Teissere and Beck (1973); 5, Patterson (1984); 6, Irving and Irving (1982), entries 39, 42–45.

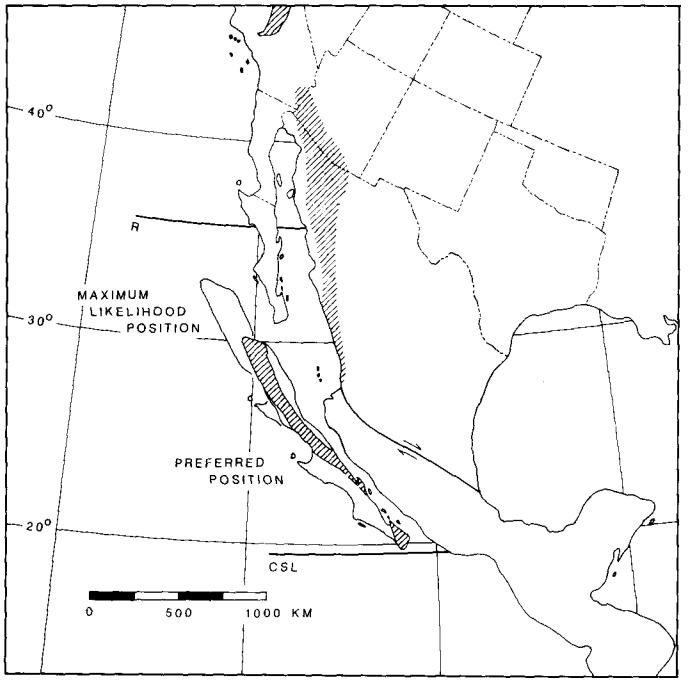


Figure 11. Paleogeographic reconstructions of peninsular California and southwestern North America for the Cretaceous. Latitudinal lines R and CSL indicate the northernmost and southernmost 95% confidence limits (after Demarest, 1983) for the northern tip (Riverside, California) and southern tip (Cabo San Lucas), respectively, of peninsular California in the maximum likelihood position. Shaded regions indicate Mesozoic plutonic rocks as in Figure 1. The strike-slip fault beneath the trans-Mexican volcanic belt is depicted as proposed by Gastil and Jensky (1973).

apparently be facilitated by removing the 435 km of right-lateral displacement along the fault beneath the trans-Mexican volcanic belt (Figs. 1 and 11), as suggested by Gastil and Jensky (1973).

Although the inferred post-Cretaceous northward movement of ~1,200 km is greater than that previously proposed for the Baja Peninsula from available geologic evidence, it remains within the constraints of relative displacement between the North American, Farallon, and Pacific plates. The tangential component of this displacement has exceeded 600 km for the past 30 m.y. and, since Cretaceous time, may have exceeded several times that amount (Atwater and Molnar, 1973; Engebretson, 1983). The required tectonic transport could be accomplished either by intermittent transform faulting along the plate margin or by continuous transform faulting east of the plate margin driven by oblique subduction (Fitch, 1972; Beck, 1980; Beck and others, 1981). Transform faulting associated with oblique subduction appears more likely because current plate-motion models (Engebretson, 1983; Page and Engebretson, 1984) indicate maximum convergence rates between the Farallon and North American plates, coincident with the subduction of older crust, during latest Cretaceous and early Tertiary time.

Completion of the pre-6 m.y. B.P. northward translation of the composite Baja California Peninsula by Miocene time is indicated by the continuity of Miocene volcanic rocks between the Baja Peninsula and the Sierra Madre Occidental of mainland Mexico (Gastil and others, 1979; Hausback, 1984; Fig. 1). Paleomagnetic results from 15–25-m.y.-old volcanic and sedimentary rocks from the Comondu and Isidro Formations of Baja California Sur (B. P. Hausback, 1984, written commun.) show no significant latitudinal displacement with respect to stable North America for the Baja Peninsula prior to the 6-m.y. B.P. opening of the Gulf of California. In addition, deceleration of the rapid early Tertiary Farallon–North America plate convergence occurred between 37 and 28 m.y. B.P. (Engebretson, 1983; Page and Engebretson, 1984), and offshore magnetic anomalies along the Middle America Trench indicate a change from right-lateral oblique to perpendicular subduction by late Miocene time (Karig and others, 1978).

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